```
clc; clear; close all;

%% IIR Filter Coefficients
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];

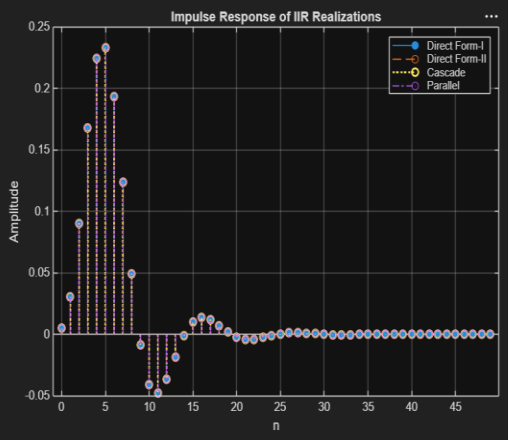
N = 50;
x = [1 zeros(1,N-1)];       % Unit impulse
n = 0:N-1;

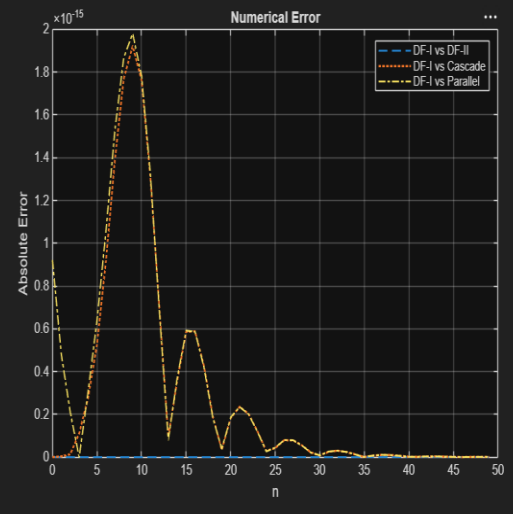
%% 1. Direct Form-I and Direct Form-II
y1 = filter(b,a,x);         % Direct Form-I
y2 = filter(b,a,x);         % Direct Form-II

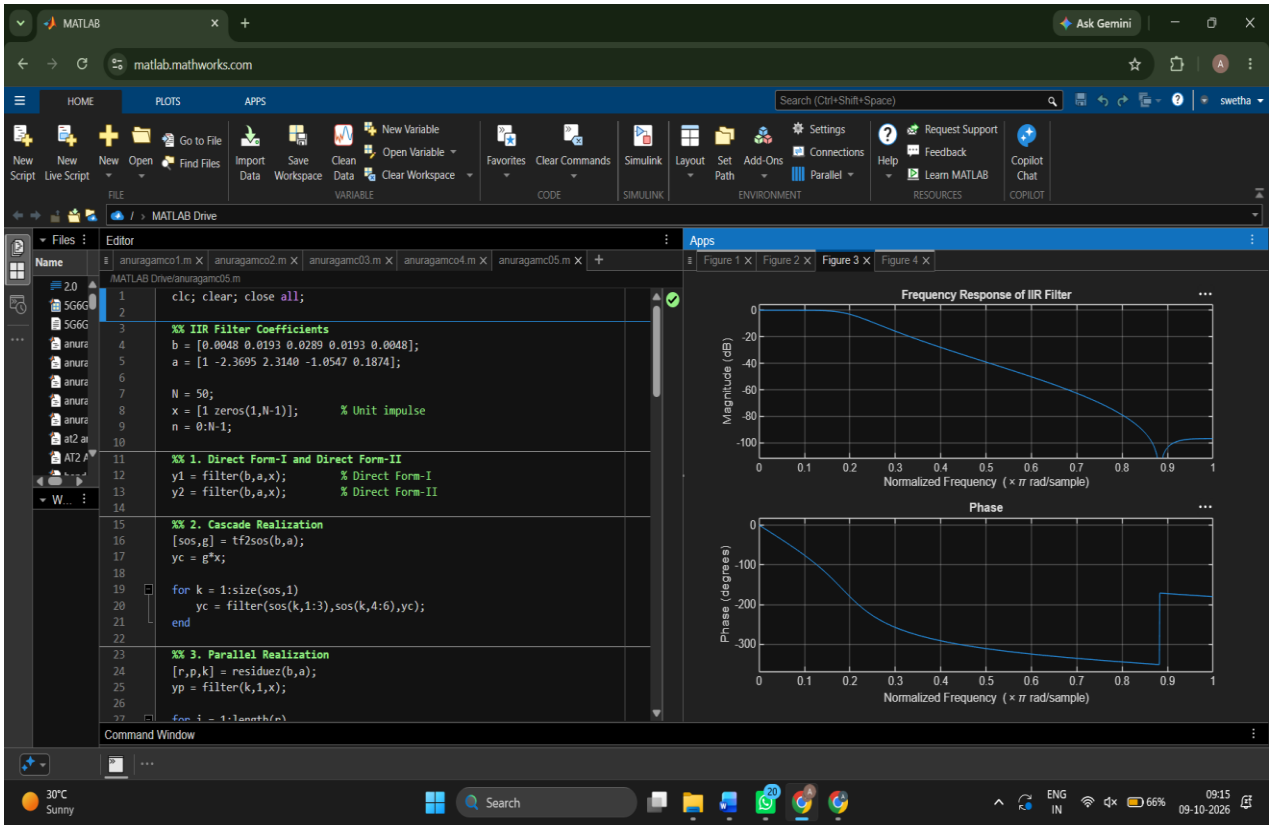
%% 2. Cascade Realization
[sos,g] = tf2sos(b,a);
yc = g*x;

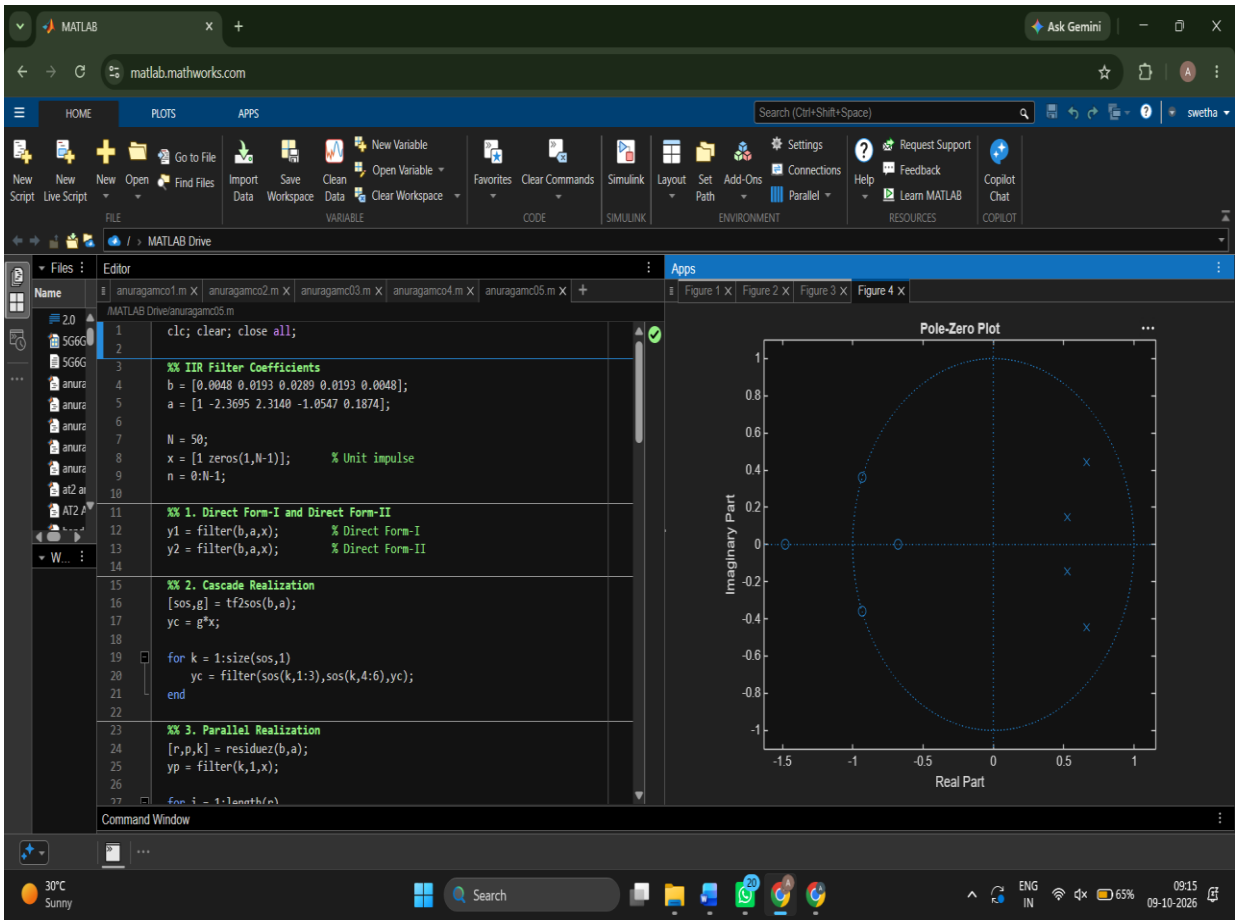
for k = 1:size(sos,1)
    yc = filter(sos(k,1:3),sos(k,4:6),yc);
end

%% 3. Parallel Realization
[r,p,k] = residuez(b,a);
yp = filter(k,1,x);
```






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end
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[r,p,k] = residuez(b,a);
yp = filter(k,1,x);
Figure 1
Figure 2
Figure 3
Figure 4
Frequency Response of IIR Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Command Window
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%% 3. Parallel Realization
[r,p,k] = residuez(b,a);
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Figure 1
Figure 2
Figure 3
Figure 4
Pole-Zero Plot
Imaginary Part
Real Part
Command Window
30°C
Sunny
Search
ENG
IN
65%
09:15
09-10-2026